

Third Normal Form (3NF)

Introduction

Third Normal Form (3NF) is a crucial stage in database normalization that removes **transitive dependencies** to reduce redundancy and anomalies.

Motivation for 3NF

- Removes **transitive dependencies** to reduce redundancy.
- BCNF is stricter but may not be **dependency preserving**.
- 3NF is a weaker form that ensures lossless join and dependency preservation.
- Guarantees functional dependencies can be checked **without joins**.

Definition

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A table is in **3NF** if:

1. It is already in **2NF**.
2. There are **no transitive dependencies** — non-key attributes depend only on the primary key.

Formal Definition with Functional Dependencies (FDs)

Formal Definition

A relation schema R is in **3NF** (w.r.t. functional dependencies F) if, for every non-trivial FD:

$$\alpha \rightarrow \beta \quad \text{in} \quad F^+$$

at least one of the following holds:

1. **Trivial FD:** $\beta \subseteq \alpha$.
2. α is a **superkey**.
3. Every attribute in $\beta - \alpha$ is contained in some **candidate key** of R .

Note: Condition 3 is weaker than BCNF — attributes may come from different candidate keys.

Example: Removing Transitive Dependency

Initial Table:

Student_ID	Student_Name	Department_ID	Department_Name
101	Alice	D1	Computer Science
102	Bob	D2	Physics
103	Charlie	D1	Computer Science

Primary Key: Student_ID

Problem: Department_Name depends on Department_ID, not directly on Student_ID

$$\text{Student_ID} \rightarrow \text{Department_ID} \rightarrow \text{Department_Name}$$

Solution: Split into two tables:

- Students(Student_ID, Student_Name, Department_ID)
- Departments(Department_ID, Department_Name)

Example: 3NF Relation That Is Not BCNF

dept_advisor Example

Relation: dept_advisor(s_ID, i_ID, dept_name)

FDs:

$$F = \{(s_ID, dept_name) \rightarrow i_ID, \quad i_ID \rightarrow dept_name\}$$

Candidate Keys: $\{s_ID, dept_name\}$ and $\{s_ID, i_ID\}$

- First FD: LHS is a superkey $\Rightarrow$ satisfies BCNF. - Second FD: i_ID is not a superkey, but $dept_name$ (RHS) is part of a candidate key $\Rightarrow$ satisfies 3NF.

Conclusion: Relation is in **3NF** but not BCNF.

Redundancy in 3NF

Redundancy in 3NF

- 3NF may still allow some repeated values.
- This can lead to minor anomalies but ensures dependency preservation.

Testing for 3NF

Testing for 3NF

1. Check only functional dependencies in F .
2. For each $\alpha \rightarrow \beta$:
 - If α is a superkey $\rightarrow$ satisfies 3NF.
 - Else, check if each attribute in $\beta - \alpha$ is part of some candidate key.

3NF Decomposition Algorithm

3NF Decomposition Algorithm

1. Compute **canonical cover** F_c for F .
2. For each FD $\alpha \rightarrow \beta$ in F_c :
 - If no schema contains $\alpha \cup \beta$, create a relation R_i with these attributes.
3. If no schema contains a candidate key, add one relation containing a candidate key.
4. Remove redundant relations.

Properties: Always produces lossless-join and dependency-preserving decomposition.

Proof: Every 3NF Relation is in 2NF

Recall definitions:

- A relation R is in **2NF** if it is in 1NF and **no non-prime attribute is partially dependent on any candidate key**. Partial dependency means an attribute depends on a *proper subset* of a candidate key.
- A relation R is in **3NF** if it is in 2NF and for every FD $\alpha \rightarrow \beta$, at least one of the following holds:
 1. α is a superkey.

2. Every attribute in β is prime (part of some candidate key).

Goal: Show that if R is in 3NF, then R is in 2NF.

Proof by contradiction:

Suppose R is in 3NF but **not** in 2NF. Then there exists a *partial dependency* violating 2NF:

$$X \rightarrow A$$

where

- X is a *proper subset* of some candidate key K (i.e., $X \subset K$),
- A is a *non-prime attribute* (i.e., A not part of any candidate key).

Since $X \subset K$, X is **not** a superkey (candidate keys are minimal superkeys). Also, A is non-prime.

Check 3NF conditions for $X \rightarrow A$:

- Is X a superkey? No.
- Is A prime? No.

Hence, $X \rightarrow A$ violates the 3NF condition.

This contradicts the assumption that R is in 3NF.

Therefore, if R is in 3NF, no partial dependency exists, so R is in 2NF.

$$\boxed{3NF \implies 2NF}$$

■

1 Examples

1.1 Example 1

Find a lossless, dependency-preserving decomposition of R into relations in 3NF

Relation schema:

$$R = (A, B, C, D, E)$$

Functional dependencies:

$$F = \{A \rightarrow BC, \quad CD \rightarrow E, \quad B \rightarrow D, \quad E \rightarrow A\}$$

Step 1: Find a minimal cover F_c of F

Check if the FDs are minimal and simplified:

- $A \rightarrow BC$ can be split into:

$$A \rightarrow B, \quad A \rightarrow C$$

- Other FDs are already single attribute on RHS.

So minimal cover:

$$F_c = \{A \rightarrow B, \quad A \rightarrow C, \quad CD \rightarrow E, \quad B \rightarrow D, \quad E \rightarrow A\}$$

Step 2: Create relation schemas for each FD in F_c

For each FD $X \rightarrow Y$ in F_c , create a relation schema $R_i = X \cup Y$:

$$\begin{cases} R_1 = (A, B) & \text{from } A \rightarrow B \\ R_2 = (A, C) & \text{from } A \rightarrow C \\ R_3 = (C, D, E) & \text{from } CD \rightarrow E \\ R_4 = (B, D) & \text{from } B \rightarrow D \\ R_5 = (E, A) & \text{from } E \rightarrow A \end{cases}$$

Step 3: Check for key coverage

A 3NF decomposition must include a relation containing a candidate key of R .

Candidate keys of R are:

- Find key candidates by closure:

Check A^+ :

$$A \xrightarrow{A \rightarrow B} \{A, B\} \xrightarrow{B \rightarrow D} \{A, B, D\} \xrightarrow{A \rightarrow C} \{A, B, C, D\}$$

But no E yet, so A^+ does not contain all attributes.

Check E^+ :

$$E \xrightarrow{E \rightarrow A} \{E, A\} \xrightarrow{A \rightarrow B} \{A, B, E\} \xrightarrow{B \rightarrow D} \{A, B, D, E\} \xrightarrow{A \rightarrow C} \{A, B, C, D, E\}$$

So E is a candidate key.

None of the relations $R_1, \dots, R_5$ contains the entire key E alone, but $R_5 = (E, A)$ contains E and A .

Since R_5 contains the key attribute E , we satisfy the condition.

Step 4: Remove redundant relations

Note that $R_2 = (A, C)$ and $R_1 = (A, B)$ can be combined into $R_6 = (A, B, C)$.

Also, $R_5 = (E, A)$ contains $E \rightarrow A$, and $R_3 = (C, D, E)$ contains $CD \rightarrow E$.

The combined set of relations for a more concise decomposition is:

$$\begin{cases} R_6 = (A, B, C) \\ R_3 = (C, D, E) \\ R_4 = (B, D) \\ R_5 = (E, A) \end{cases}$$

Step 5: Verify lossless join and dependency preservation

Lossless join:

- Relation R_6 and R_3 share attribute C . - Relation R_3 and R_5 share attribute E .
- Relation R_6 and R_5 share attribute A . - Relation R_4 shares attribute B with R_6 .
Using the fact that each FD is represented and the keys are covered, the decomposition is lossless.

Dependency preservation:

All original FDs in F are represented in the relations:

- $A \rightarrow BC$ in R_6 . - $CD \rightarrow E$ in R_3 . - $B \rightarrow D$ in R_4 . - $E \rightarrow A$ in R_5 .

Final 3NF Decomposition

$$\begin{cases} R_6 = (A, B, C) & A \rightarrow B, A \rightarrow C \\ R_3 = (C, D, E) & CD \rightarrow E \\ R_4 = (B, D) & B \rightarrow D \\ R_5 = (E, A) & E \rightarrow A \end{cases}$$

All relations are in 3NF, the decomposition is lossless, and dependencies are preserved.

Summary Table

3NF Summary

Rule	Description
Already in 2NF	Table must satisfy all 2NF rules.
No Transitive Dependencies	Non-key attributes must depend only on the primary key, not on other non-key attributes.
Superkey or Key Attribute Rule	For each FD, the RHS attribute must be part of a candidate key if LHS is not a superkey.